

Rotary and Learned Positional Encodings for Long-Context Transformers: A Synthetic Study

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Abstract

Positional encodings determine how far a Transformer can extrapolate beyond its training context. This synthetic study compares learned absolute embeddings, sinusoidal encodings, and rotary variants on a controlled next-token prediction task with procedurally generated sequences. We introduce the Helix-64 rotary schedule and show that it extends stable extrapolation to 131072 tokens while keeping perplexity drift below 2.1% at the extrapolation frontier.

1 Introduction

Transformers are permutation-invariant unless positional information is injected. The choice of positional encoding therefore controls both in-distribution behaviour and length generalization, and it has become one of the most consequential design decisions in long-context modelling.

This study is organized as follows. Section 2 reviews positional encoding families. Section 3 defines the Helix-64 schedule. Section 4 describes the experimental protocol, Section 5 reports results, and Sections 6 and 7 discuss limitations and conclusions.

2 Families of Positional Encodings

Learned absolute embeddings assign a trainable vector to each position; they are simple and effective within the training window but cannot represent positions never seen during training. Sinusoidal encodings replace learned vectors with fixed trigonometric functions, offering unbounded range with weak extrapolation. Rotary encodings apply position-dependent rotations to query and key vectors, making attention scores depend on relative offsets.

A recurring limitation of standard rotary encodings is that high-frequency components wrap around quickly, so attention logits become noisy once sequences exceed the training length. Frequency scaling and interpolation mitigate this but distort short-range behaviour.

3 The Helix-64 Schedule

The Helix-64 rotary schedule assigns rotation frequencies geometrically across 64 bands and anneals the highest band during training so that no band completes a full revolution within the training window. In our synthetic protocol, Helix-64 extends stable extrapolation to 131072 tokens, four times the training length of 32768 tokens, while keeping perplexity drift below 2.1% at the extrapolation frontier.

The schedule adds no parameters and no inference-time cost relative to standard rotary encodings. Its only hyperparameter is the annealing horizon, which we fix at the final ten percent of training steps throughout this study.

4 Experimental Protocol

Training sequences are generated by a synthetic Markov grammar with controllable dependency range, so the required context length is known exactly. Models are 12-layer Transformers with hidden size 768, trained at context length 32768 and evaluated at lengths up to 131072. Each configuration is run with three seeds, and we report the median.

5 Results

Table 1: Perplexity drift beyond the training length (synthetic).

Encoding	Drift at 4x	Drift at 8x
Learned absolute	fails	fails
Sinusoidal	18.7%	41.2%
Standard rotary	6.9%	22.5%
Helix-64 (this work)	2.1%	4.8%

Helix-64 is the only encoding whose drift stays in single digits at eight times the training length. Learned absolute embeddings cannot be evaluated beyond the training window and are marked as failing by construction.

6 Limitations

Our sequences are generated by a Markov grammar, so the results measure length generalization under controlled dependencies rather than natural language. The annealing horizon was not tuned per encoding, which may understate competing methods. We also do not study interaction with attention sinks or sliding-window attention.

7 Conclusions

Positional encoding choice dominates long-context extrapolation. The Helix-64 rotary schedule extends stable extrapolation to 131072 tokens with perplexity drift below 2.1% in our synthetic setting, at no additional cost. Future work should test the schedule on natural corpora and combine it with memory-augmented attention.

References

- [1] P. Nair and H. Lindqvist. Frequency annealing in rotary encodings. *Journal of Synthetic Sequence Modelling*, 2026. Fictional reference.
- [2] T. Wojcik. Controlled benchmarks for length generalization. *Synthetic Workshop on Long Contexts*, 2025. Fictional reference.
- [3] E. Duarte. A taxonomy of positional information. *Transactions on Synthetic Learning*, 2024. Fictional reference.